

DeepSeek-V2

A Strong, Economical, and Efficient MoE
Language Model

236B Total / 21B Activated Params

128K Context

Based on: [arXiv:2405.04434](https://arxiv.org/abs/2405.04434)

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

The Goal: Higher Intelligence

Scaling up model size (parameters) consistently yields better reasoning, knowledge, and capabilities.



The Bottlenecks

1. Training Costs

Dense models require activating all parameters for every token, leading to prohibitive compute costs.

2. Inference Speed (KV Cache)

Autoregressive generation is bottlenecked by the size of the Key-Value (KV) cache stored in memory.

DeepSeek-V2 Architecture: MLA + ~~DeepSeekMoE~~

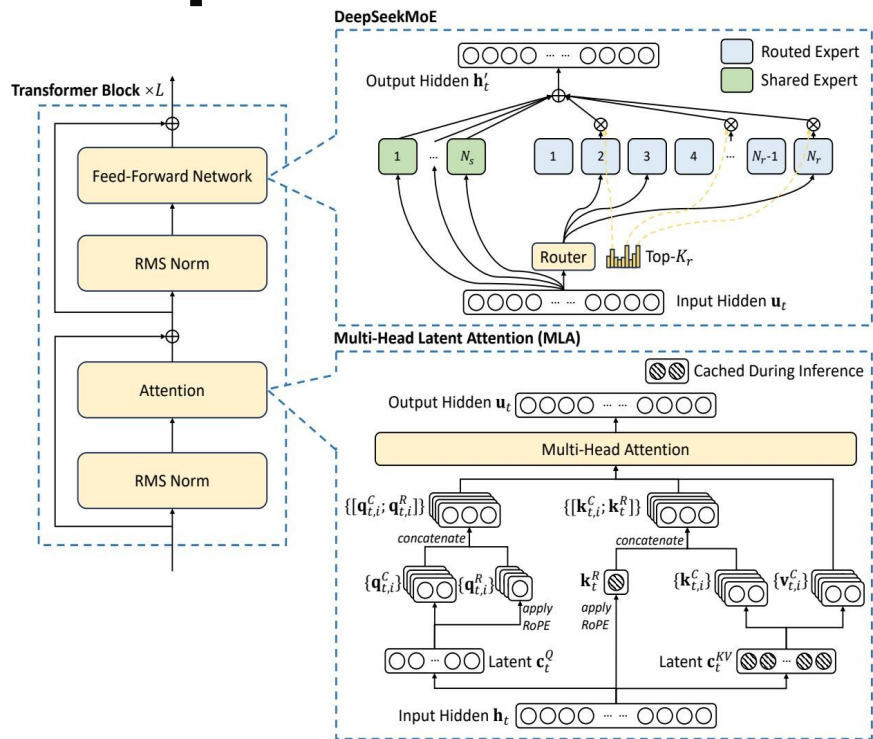


Fig 1: DeepSeek-V2 Transformer Block Architecture

Innovating the Transformer Block

1. Multi-Head Latent Attention (MLA)

- Replaces standard Multi-Head Attention.
- Compresses the KV cache to solve inference bottlenecks while maintaining performance.

2. DeepSeekMoE

- Replaces dense Feed-Forward Networks (FFN).
- Uses highly sparse routing (activating only 21B out of 236B parameters per token) to drastically reduce training costs.

Multi-Head Latent Attention Compresses KV

Cache by 93.3%

Conventional Multi-Head Attention (MHA) stores full key-value pairs per head. Alternatives like GQA/MQA reduce cache but sacrifice performance.

The MLA Solution

MLA introduces low-rank key-value joint compression. Instead of caching full vectors, it caches a compact latent vector.

Decoupled RoPE for positional encoding

On-the-fly reconstruction via learned projection matrices

Key Mechanisms:

Performance exceeds MHA with drastically less cache

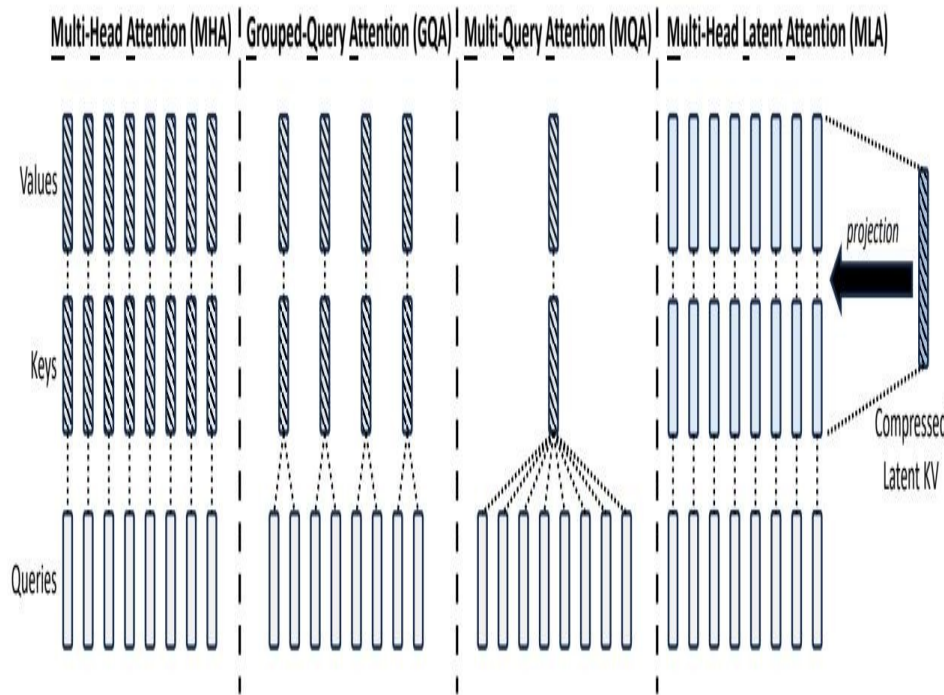


Fig 2: Attention mechanism comparison (MHA vs GQA vs MQA vs MLA)

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Two Expert Types

Shared experts (always active for common knowledge) and Routed experts (sparsely activated via top-K).

Fine-Grained Segmentation

Increases specialization potential by having many smaller experts rather than a few large ones.

Device-Limited Routing

Constrains tokens to experts on at most M devices, significantly reducing cross-node communication overhead.

Load Balancing

Auxiliary losses balance load across experts and devices. Token-dropping prevents overflow during training.

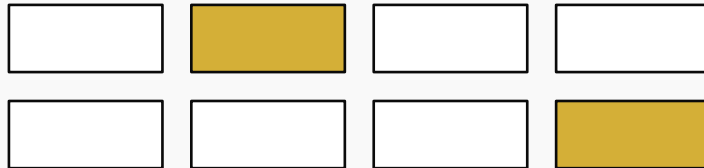
DeepSeekMoE Routing Mechanism

Shared Experts (Always Active)



Top-Kr
Router

Routed Experts (Sparsely Activated)



Efficiency Gains: DeepSeek-V2 vs. DeepSeek 67B

Relative Improvements over Dense Predecessor

Training Cost

↓ **42.5%**

Achieved via
DeepSeekMoE's sparse
activation

KV Cache Size

↓ **93.3%**

Achieved via Multi-Head
Latent Attention
compression

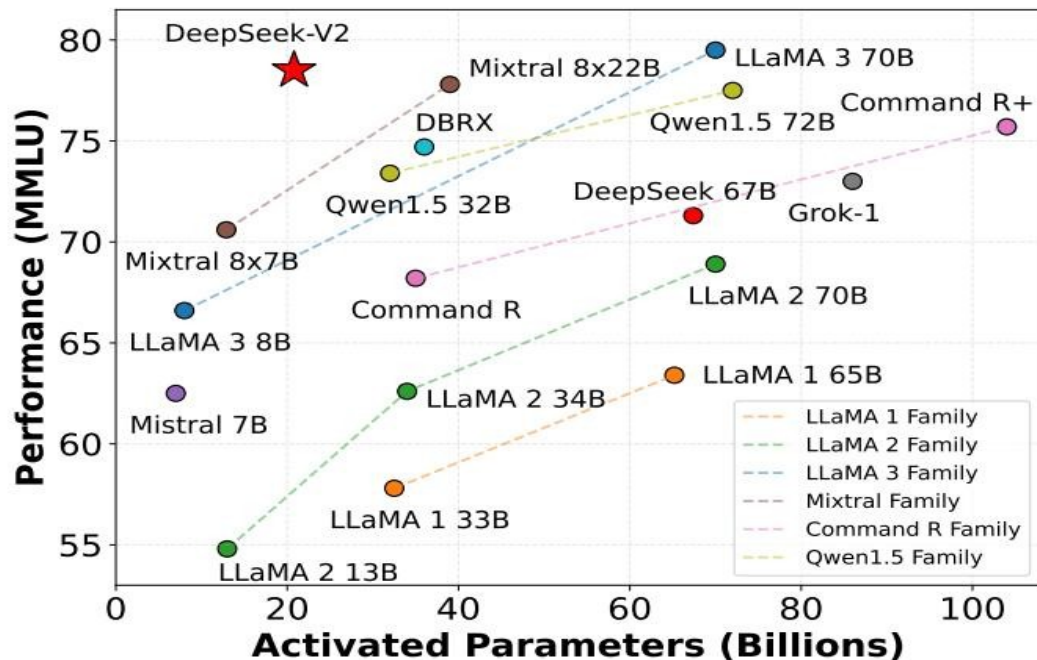
Max Throughput

↑ **5.76×**

Enables practical, large-
scale deployment

DeepSeek-V2 Tops MMLU Among Open-Source Models

Pareto-Optimal Efficiency at 21B Activated Parameters



Model Comparison

Model	Act. Params	MMLU
DeepSeek-V2	21B	78.5
Mixtral 8x22B	39B	77.6
LLaMA 3 70B	70B	77.9
Qwen1.5 72B	72B	75.6
DeepSeek 67B	67B	71.5
DeepSeek-V2 outperforms large dense models (LLaMA 3 70B, Qwen1.5 72B) while activating significantly fewer parameters per token.		

Alignment: DeepSeek-V2 Chat (RL) Rivals Closed-Source

Alignment Pipeline

1

Supervised Fine-Tuning (SFT)

Trained on 1.5M high-quality conversational sessions.



2

Reinforcement Learning

Optimized via Group Relative Policy Optimization (GRPO).

Open-Ended Conversation Benchmarks

AlpacaEval 2.0

38.9%

MT-Bench
(Overall)

8.97

AlignBench
(Chinese)

7.91

Outperforms all open-source models and most closed-source models on Chinese AlignBench.

KEY TAKEAWAYS

Architectural Innovation Unlocks Efficient Scaling

- ✓ **MLA solves the KV cache bottleneck**, outperforming MHA while compressing cache by 93.3%.
 - ✓ **DeepSeekMoE enables economical training**, reducing costs by 42.5% via fine-grained experts.
 - ✓ **Pareto-optimal performance**: Matches 70B dense models on MMLU with only 21B activated parameters.
 - ✓ **Strong multilingual alignment** makes DeepSeek-V2 a leading open-source chat model.
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